

3(a) The increase in the internal energy of a system is the sum of the heat supplied to the system and the work done on the system.

(b)(i) Apply the ideal gas law $pV = nRT$. Since pressure p is constant, $V \propto T$

$$\frac{T_i}{T_f} = \frac{V_i}{V_f}$$

$$\frac{T_i}{(273.15 + 105)} = \frac{2200}{2800}$$

$$T_i = 297.118 \text{ K} = 23.97^\circ\text{C}$$

$$= \mathbf{24.0^\circ\text{C}}$$

(ii) Work done by the expanding balloon on the atmosphere,

$$W_{\text{by}} = p\Delta V = (1.01 \times 10^5)(2800 - 2200)$$

$$= \mathbf{6.06 \times 10^7 \text{ J}}$$

$$= 60.6 \text{ MJ}$$

(iii) For an ideal gas, its internal energy U is directly proportional to its thermodynamic temperature T , i.e. $U \propto T$.

Since the temperature has increased after the heating, the internal energy will also increase.

(iv) Apply 1st law of thermodynamics:

$$\Delta U = Q + W_{\text{on}} = Q - W_{\text{by}}$$

$$116 \text{ MJ} = Q - 60.6 \text{ MJ}$$

$$\therefore \text{heat supplied, } Q = 176.6 \text{ MJ}$$

Minimum mass of propane required

$$= 176.6 \div 50 = \mathbf{3.53 \text{ kg}}$$